

Team Name	Team AURA
University	University of Peradeniya (Faculty of Engineering)
Product Name	AURA (Automated Urban Restaurant Assistant)
Product Category	Smart Hospitality Robotics & Restaurant Assistant , IoT

EXECUTIVE SUMMARY

Project Introduction

AURA (Automated Urban Restaurant Assistant) is an end-to-end intelligent tabletop robotic dining assistant and cloud-integrated restaurant operations platform designed to modernize the dine-in food service industry. Developed by Team ReLife from the Faculty of Engineering, University of Peradeniya, AURA bridges physical robotic automation with a robust full-stack software ecosystem to solve persistent operational bottlenecks in commercial dining spaces.

By replacing traditional manual order-taking with a table-side interactive robotic assistant featuring voice AI, interactive touchscreen capabilities, real-time kitchen dispatching, and PayHere gateway payment integration, AURA streamlines the entire customer journey from table arrival to bill settlement. Operating on a reliable plug-and-play direct current (AC adapter) power architecture, AURA delivers high-end interactive guest experiences without the exorbitant costs or spatial footprint of heavy walking service robots.

Key Highlights

- **Hardware Innovation:** Interactive Tabletop Robotic Hardware: Compact, plug-and-play tabletop footprint powered by a Raspberry Pi 4B controller, dual OLED digital eyes, interactive touch display, custom 3D-printed chassis, and a precision servo/gear rotation assembly.
 - **Computer Vision & Kinematics:** Vision-Guided Neck & Body Tracking: Utilizes onboard Pi Camera V2 with Computer Vision (OpenCV Haar Cascade) to track customer faces and actuate dual SG90 servos and a custom mechanical gear system, physically turning the robot to face seated guests.
 - **AI Engagement:** Conversational Voice AI Agent: Integrated with Google Gemini AI and audio modules, allowing natural language dining interactions, menu inquiries, and hands-free order requests.
 - **Full-Stack Software:** Multi-Theme Guest Interface & Web Platform: Features a dual Light/Dark theme tabletop interface, dedicated staff login portal, live kitchen display system (KDS), and a web application accessible at <https://www.aurarestaurant.tech>.
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- **Frictionless Billing:** Table-Side Payment Integration: Supports instant card and cash payment processing via the PayHere payment gateway directly on the robot interface, automatically updating kitchen and administrative dashboards upon successful settlement.
- **Guest Experience:** In-Kitchen & Table-Side Entertainment: Integrated background music, interactive mini-games, and a dedicated 'Selfie Mode' to keep guests entertained during order preparation.

PROBLEM DEFINITION AND PROPOSED SOLUTION OVERVIEW

Identifying the Problem and Solution Approach

Urban dining establishments face severe operational bottlenecks during peak hours that compromise customer satisfaction and erode profit margins. Key challenges identified through industry surveys and hospitality metrics include:

- **Long Waiting Times:** Customers endure prolonged delays waiting for waitstaff to bring menus, record orders, or present final bills, leading to decreased table turnover.
- **Order Errors & Miscommunication:** Manual order taking in noisy environments leads to misheard customization, incorrect ticket items, and food wastage.
- **Staff Shortages & High Labor Overhead:** High turnover rates and rising labor costs make maintaining adequate waitstaff increasingly difficult and financially unsustainable for mid-tier restaurants.

High Cost of Walking Robots: Existing commercial walking service robots cost over (Entry-Level AI Ordering Robots) **Cost: Rs 1,175,000 – Rs 1,510,000** (\$3,500 - \$4,500) per unit, require large physical aisle clearance, physical aisle clearance pose safety hazards in crowded spaces.

AURA solves these challenges by introducing a stationary, highly capable tabletop robotic assistant per dining table. Rather than navigating floor spaces, AURA resides on the table, offering instant, error-free interactive ordering, entertainment, and digital payments directly connected to the kitchen via low-latency AWS cloud architecture.

Problem Scope

The project scope focuses on small to medium-sized urban restaurants, casual dining establishments, themed cafés, and quick-service diners. The target ecosystem addresses both customer-facing interaction and backend operational synchronization:

- **Customer Domain Scope:** Customers receive instantaneous menu access, AI-guided voice assistance, entertaining mini-games while waiting, and convenient table-side settlement.
 - **Kitchen & Operational Scope:** Kitchen staff receive structured order tickets in real-time on a digital display, eliminating lost paper slips and kitchen-front communication delays.
 - **Management Scope:** Restaurant management gains centralized control over menu management, active table statuses, transaction ledgers, and operational analytics.
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PRODUCT OVERVIEW AND DISTINCTIVE FEATURES

Product Concept

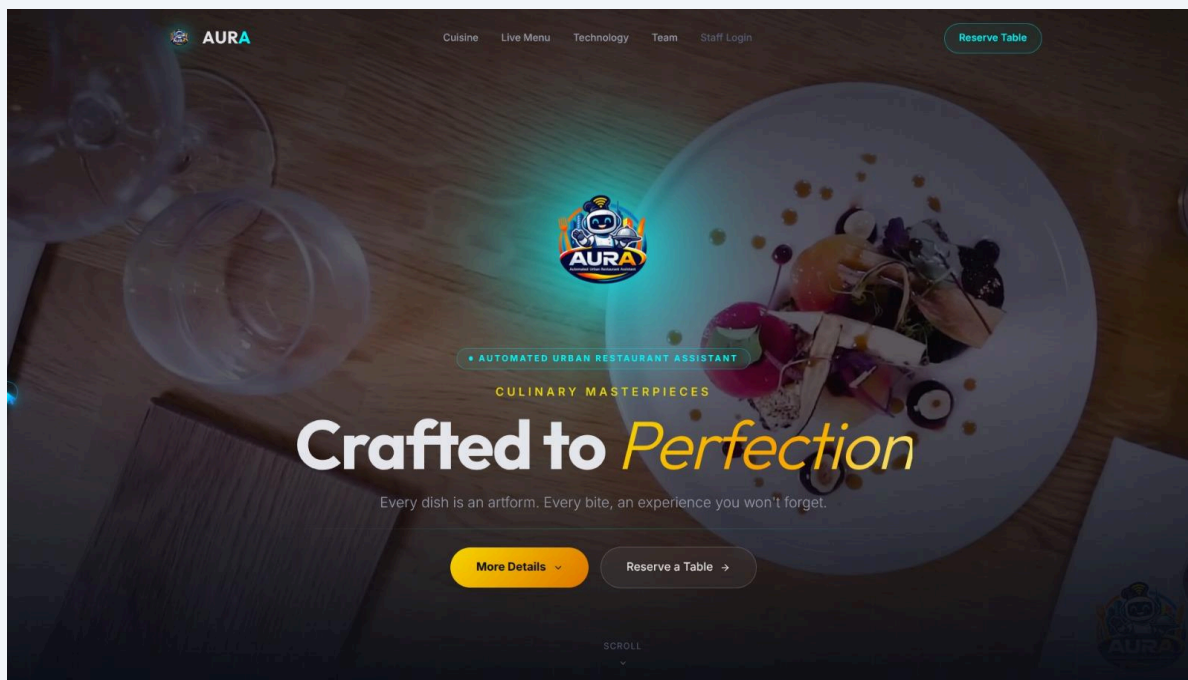
AURA is an intelligent, plug-and-play tabletop robot that serves as an active dining companion and point-of-sale terminal. Blending responsive mechanical robotics, Computer Vision face tracking, AI voice intelligence, and full-stack cloud management, AURA elevates dining into an engaging, efficient, and interactive experience.

Detailed Product Description & Workflow

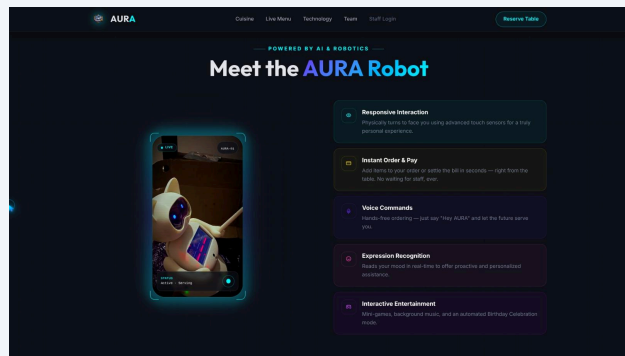
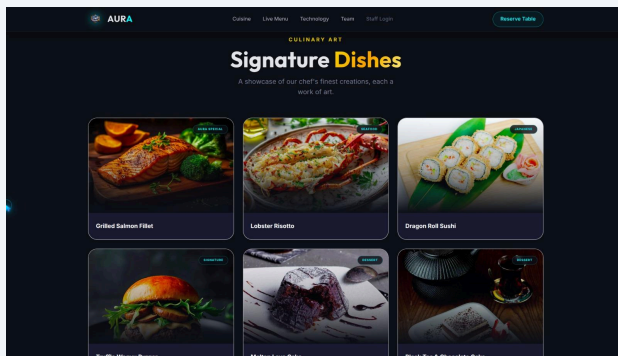
The AURA product ecosystem encompasses physical robotic units on tables, web-based guest interfaces, staff portals, live kitchen display systems (KDS), and secure cloud infrastructure. Below is a detailed walk-through of the complete product workflow accompanied by architectural and UI visual placeholders:

AURA Official Landing Page & Web Ecosystem

Figure 1: Official landing page and web platform for AURA accessible at where you can reserve you table

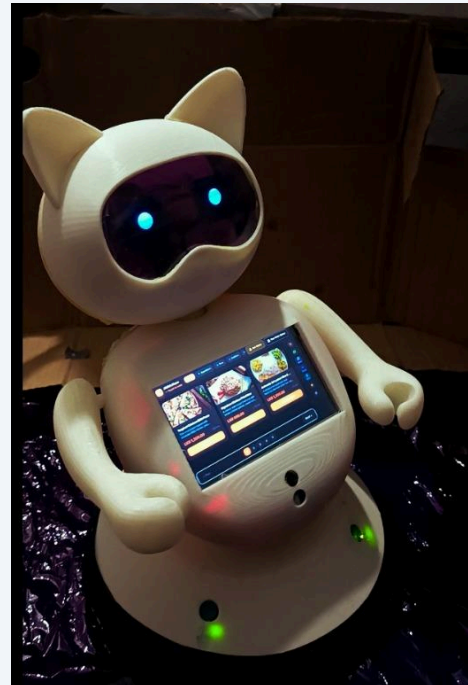
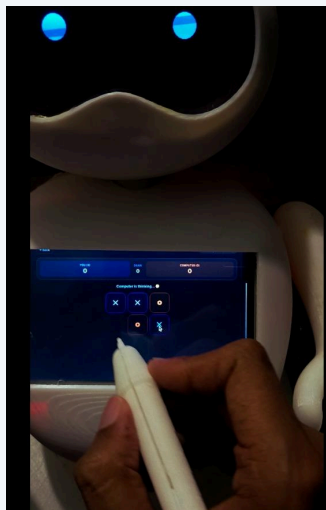
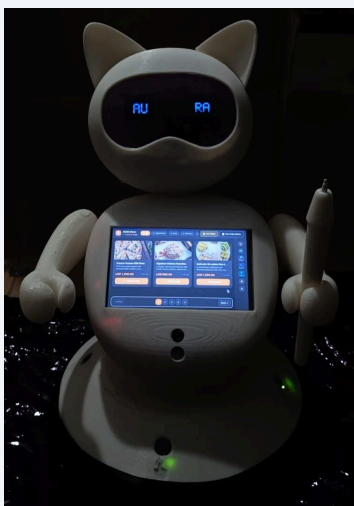
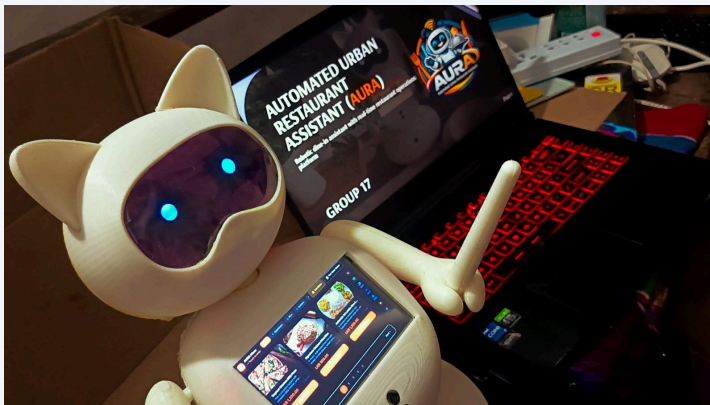


<https://www.aurarestaurant.tech>



AURA Final Physical Product – Complete Robotic Assembly

Figure 2: Final physical AURA tabletop robot fully assembled with touchscreen, dual OLED eyes, and sleek 3D-printed chassis. This will rotate where you sit when touch the sensors



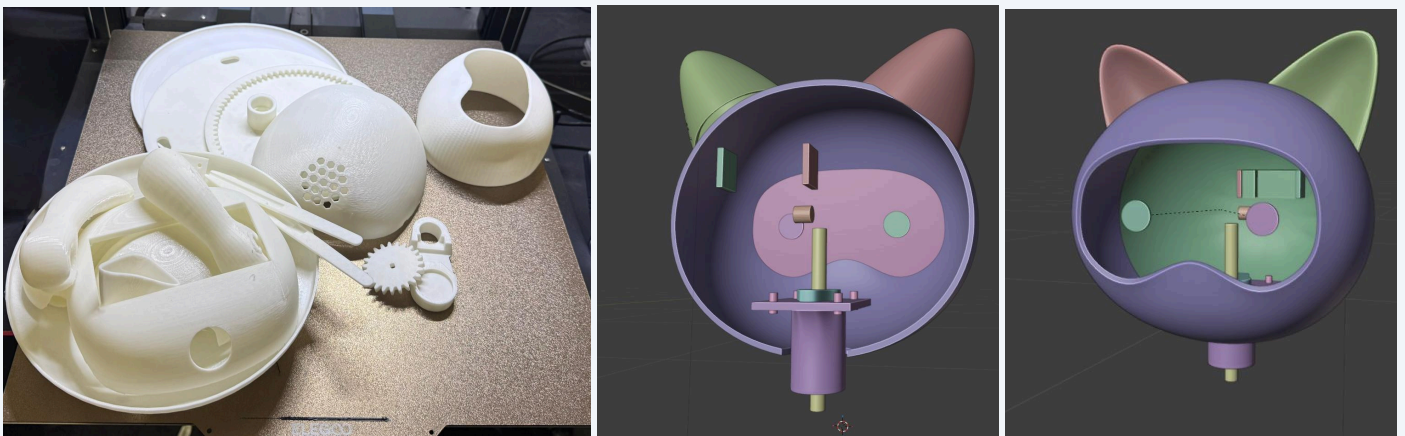
Detailed Design & Manufacturing Phase Walkthrough

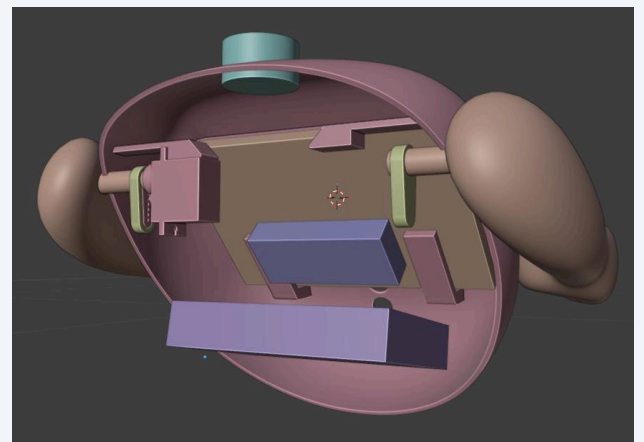
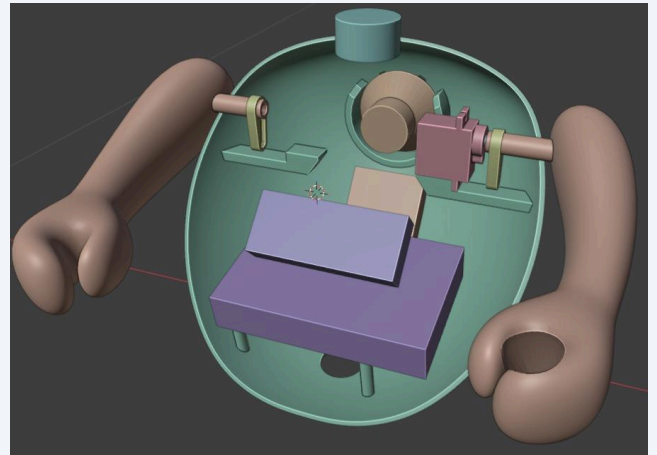
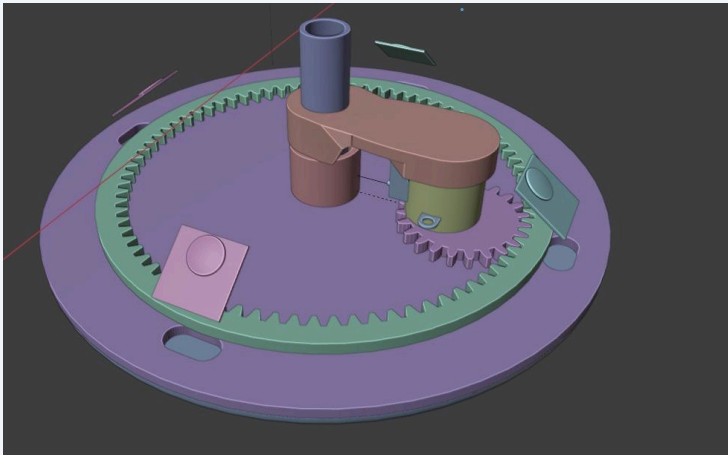
The physical construction of AURA involved iterative engineering across 3D modeling, chassis design, mechanical drive integration, packaging design, and user documentation:

- **Blender 3D Modeling & CAD:** Custom 3D modeling of the robotic head, neck assembly, base enclosure, and dual OLED eye sockets designed in Blender and CAD software to achieve an approachable, high-tech aesthetic.
- **3D Printing & Fabrication:** Precision additive manufacturing using high-strength PLA/PETG filaments, optimized for thermal stability, internal component housing, and clean wire routing.
- **Chassis & Internal Structural Frame:** Encloses the Raspberry Pi 4B, power conversion boards, audio amplifier, TTP223 touch sensors, and dual SG90 pan-tilt servo mechanisms.
- **Mechanical Gear Rotation System:** A robust custom mechanical gear system driven by stepper/servo motors that rotates the upper robotic body around the base table mount to directly face seated customers.
- **Packaging, User Manual & Sticker Pack:** Professional eco-friendly product packaging, comprehensive physical user manual for restaurant operators, and branded sticker packs for customer engagement.

Design Phase Walkthrough – 3D Printing, Chassis & Blender Design

Figure 3: Iterative design phases including Blender 3D computer models, CAD structural schematics, 3D printing, and chassis assembly.

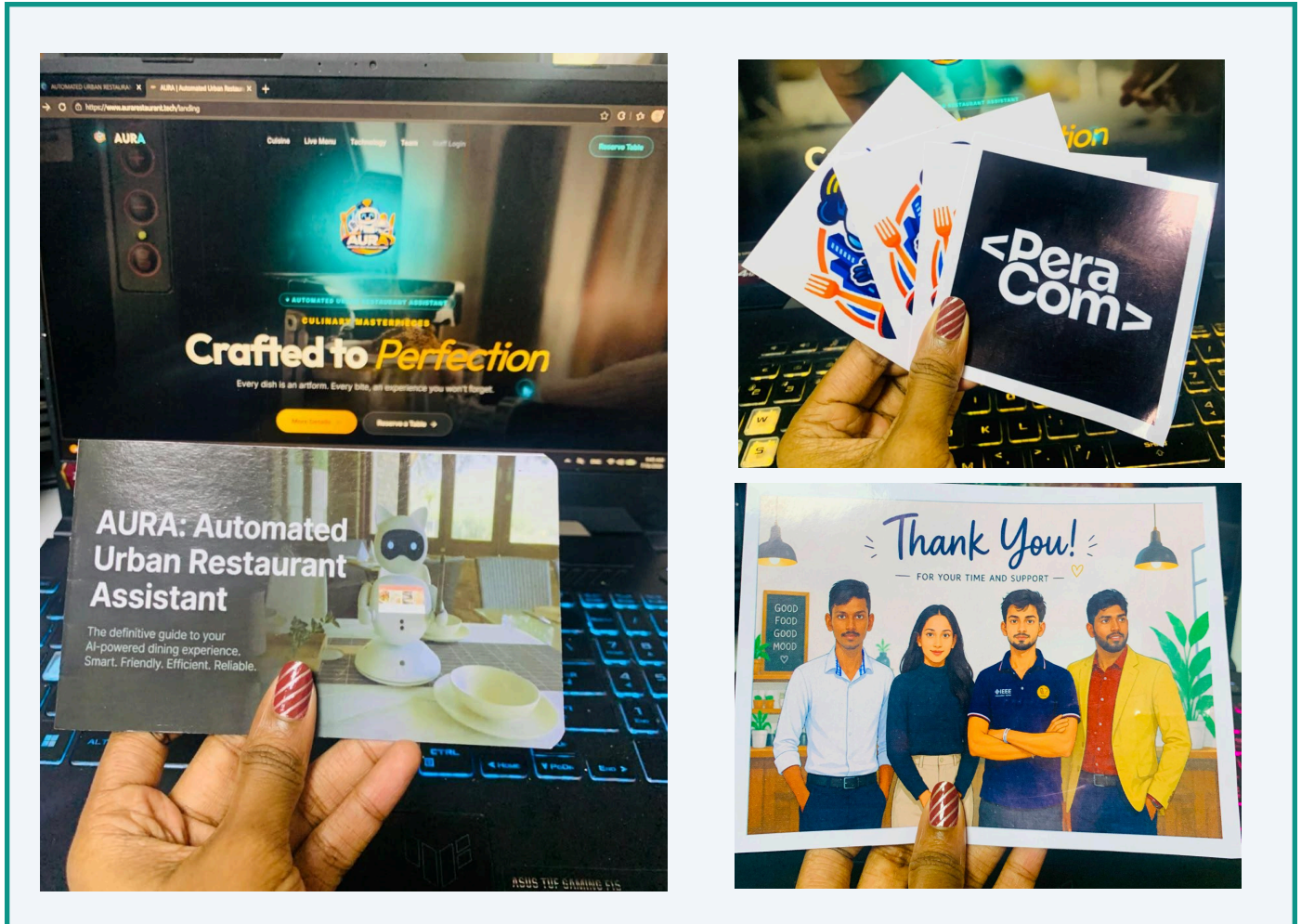




Product Packaging, User Manual & Branded Sticker Pack

Figure 4: Complete AURA packaging design, printed user manual for restaurant staff, and custom promotional sticker pack.



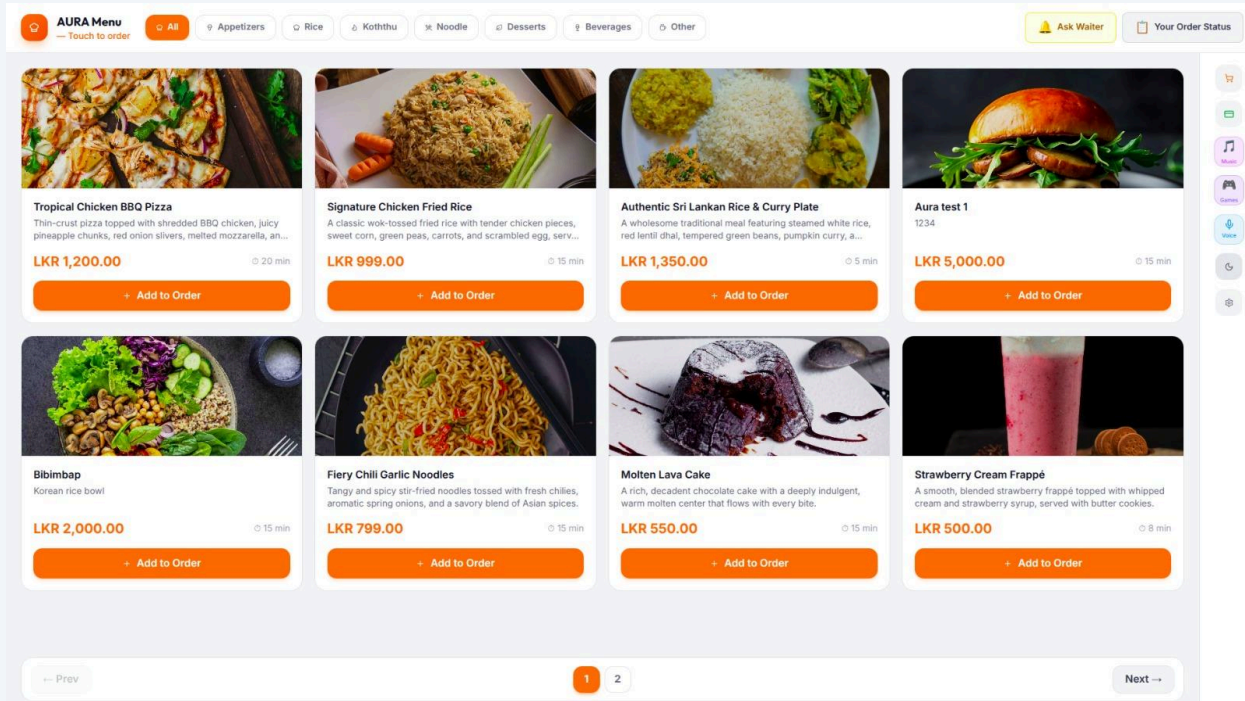
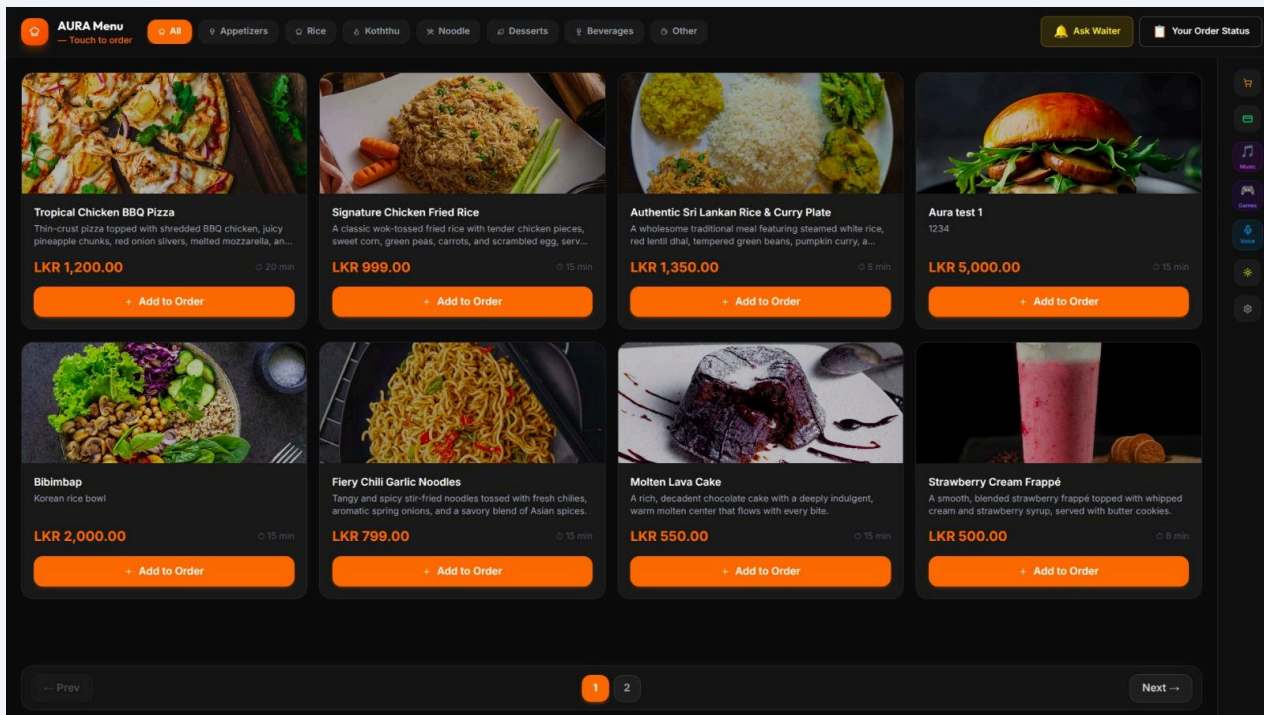


Interactive Tabletop UI & Customer Experience Workflow

Once seated, guests interact with AURA through a highly polished visual interface and natural voice commands:

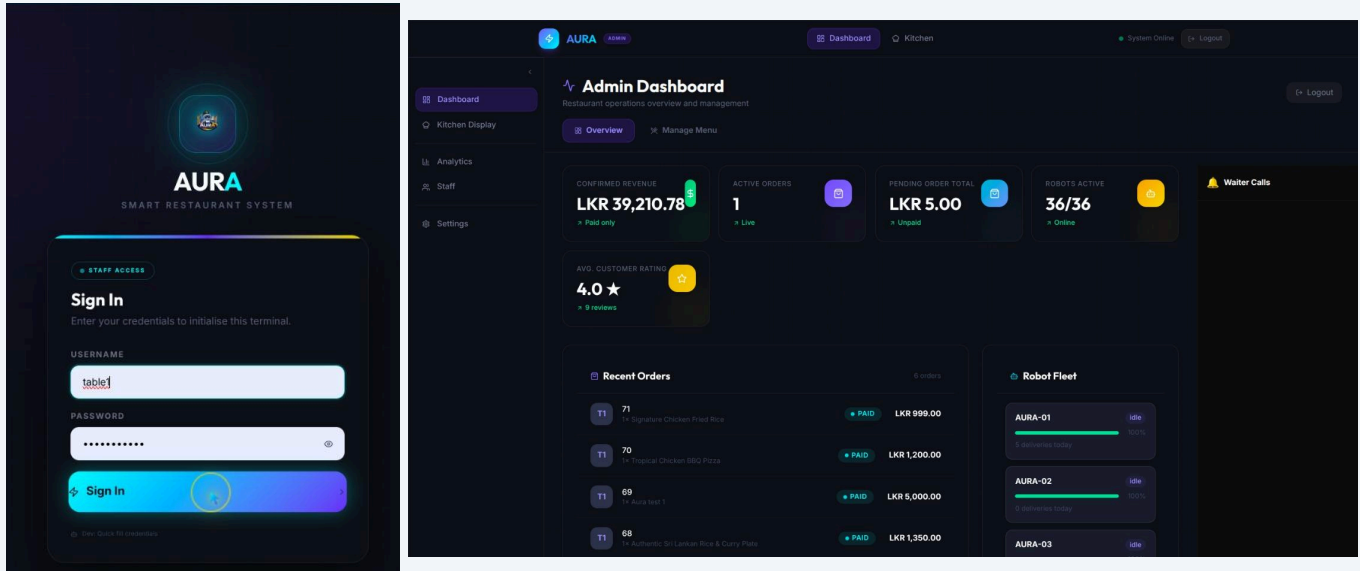
AURA Tabletop Robot Interface – Light and Dark Theme Options

Figure 5: High-contrast, responsive user interface on AURA's touchscreen displaying both Light and Dark theme modes.



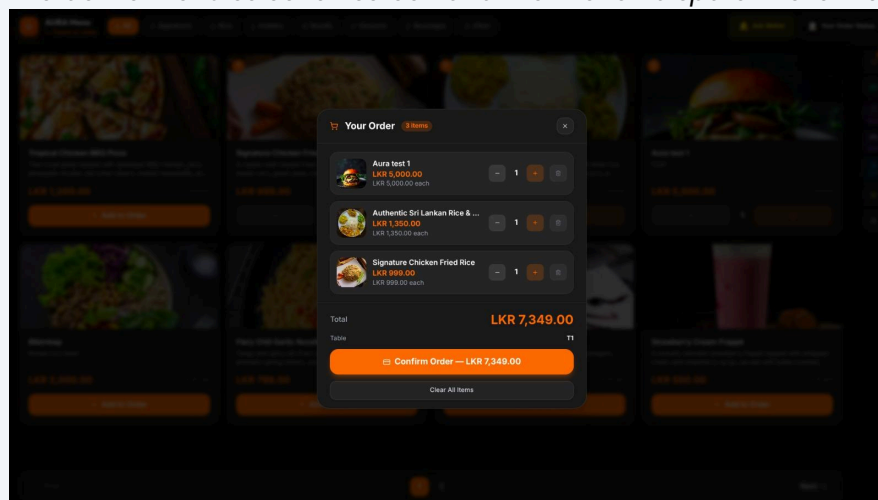
AURA Staff Login & Management Interface

Figure 6: Secure staff login portal allowing restaurant employees to manage table allocations, system settings, and menu availability.

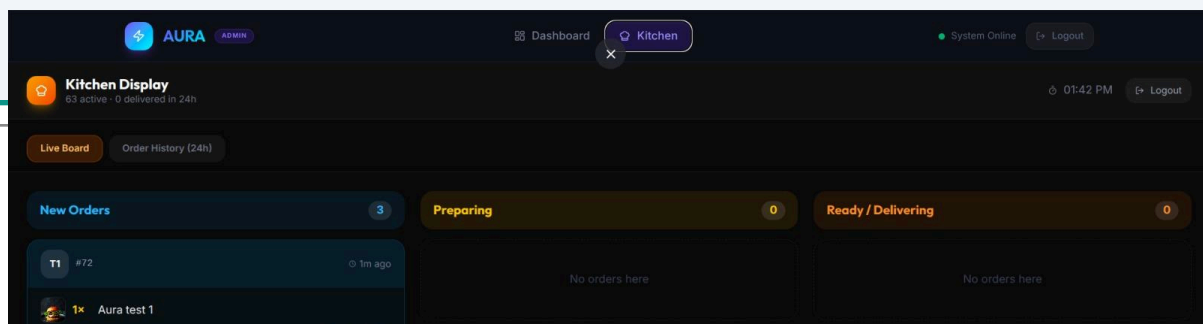


Food Ordering Workflow & Real-Time Kitchen Notification

Figure 7: Interactive menu selection screen and live kitchen dispatch ticket notification.



This is the interface that can be seen in the ROBOT Touch Screen (Instant Menu Ordering and so on)



Kitchen update instantly when order confirmed

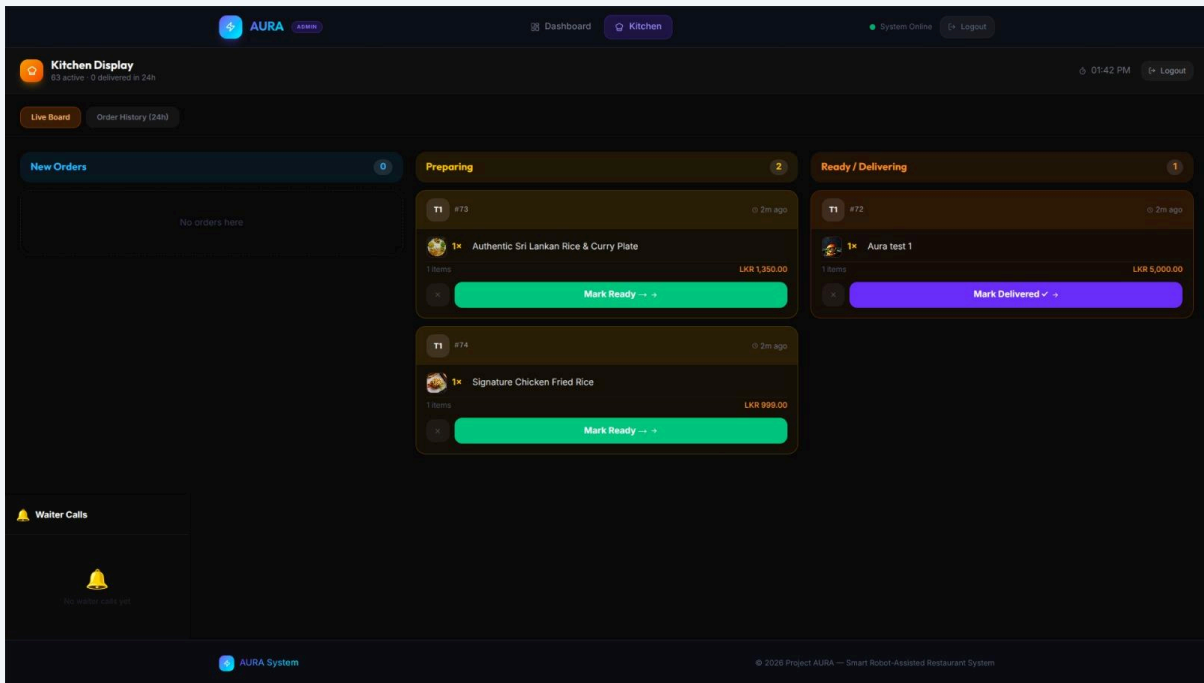
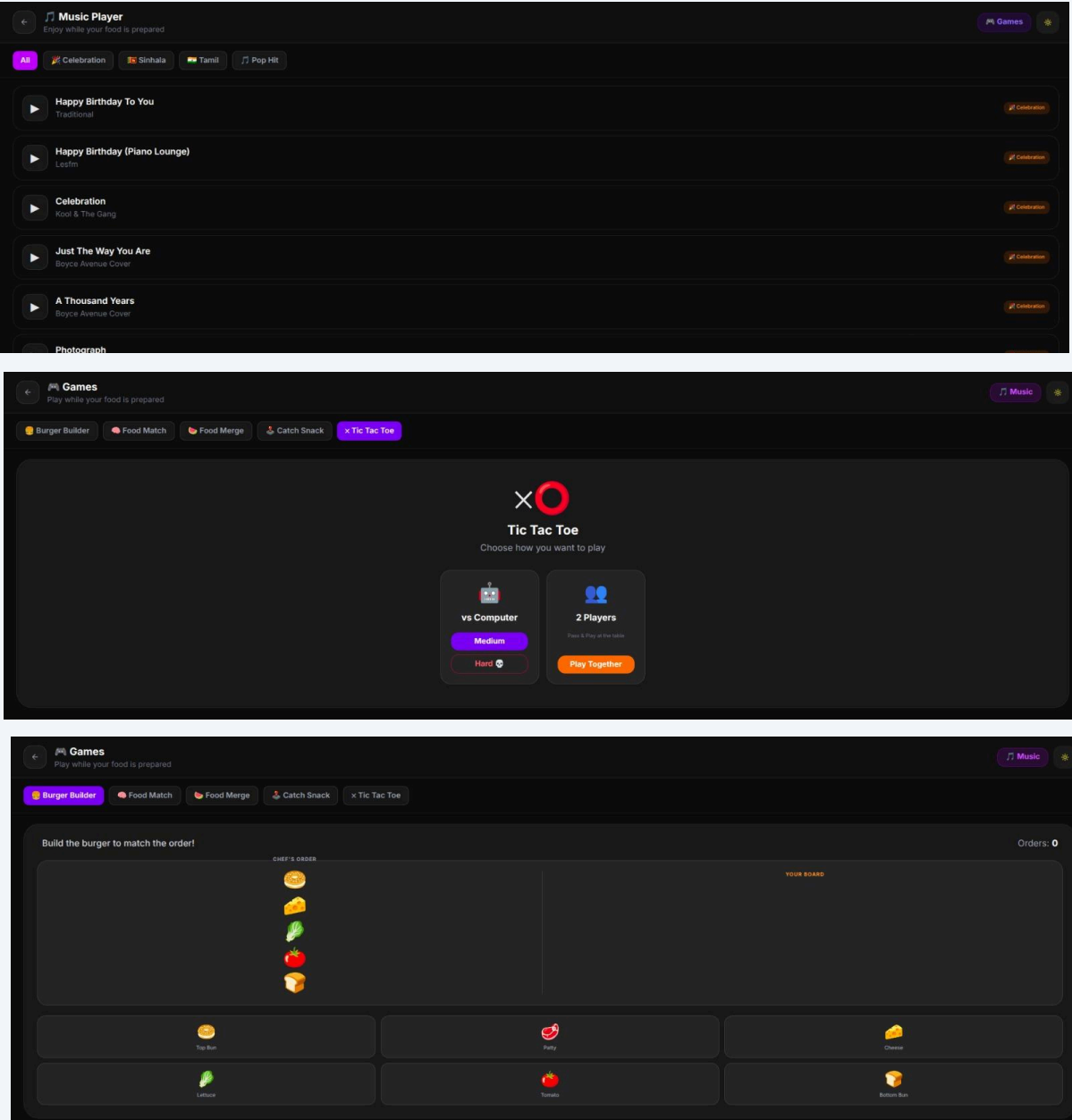


Table-Side Entertainment Module – Music Player & Mini-Games*Figure 8: Built-in entertainment suite allowing guests to listen to background music and play games while food is being prepared.*

These are in the ROBOT Touch display interface



AURA Voice AI Assistant & Cute Selfie Experience

Figure 9: Conversational AI voice interaction and interactive you can take a Selfies with our robots'

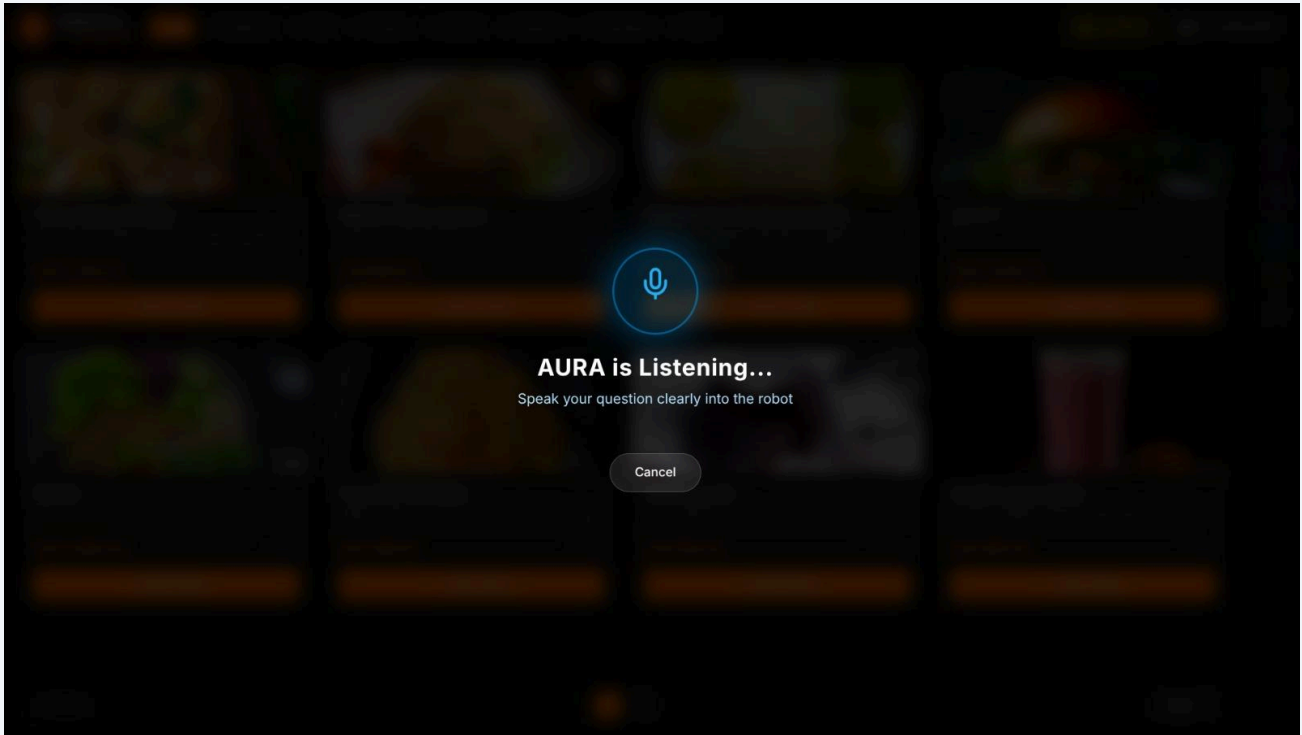
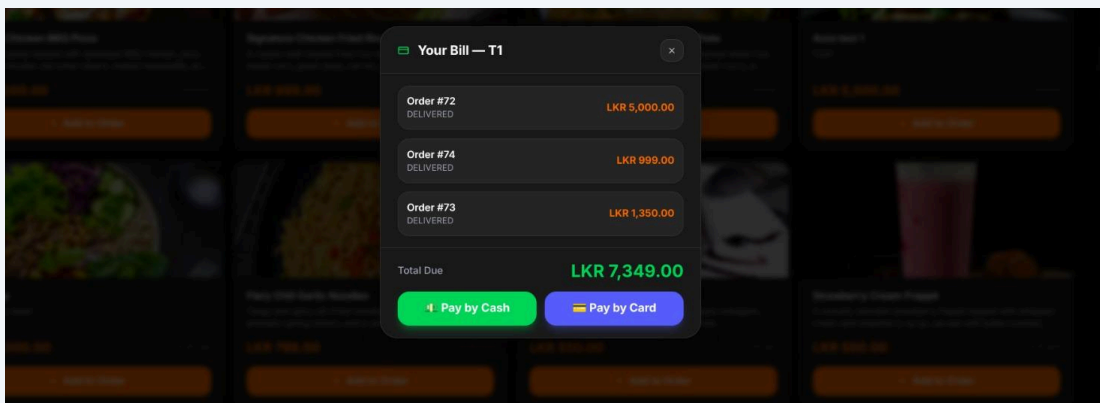
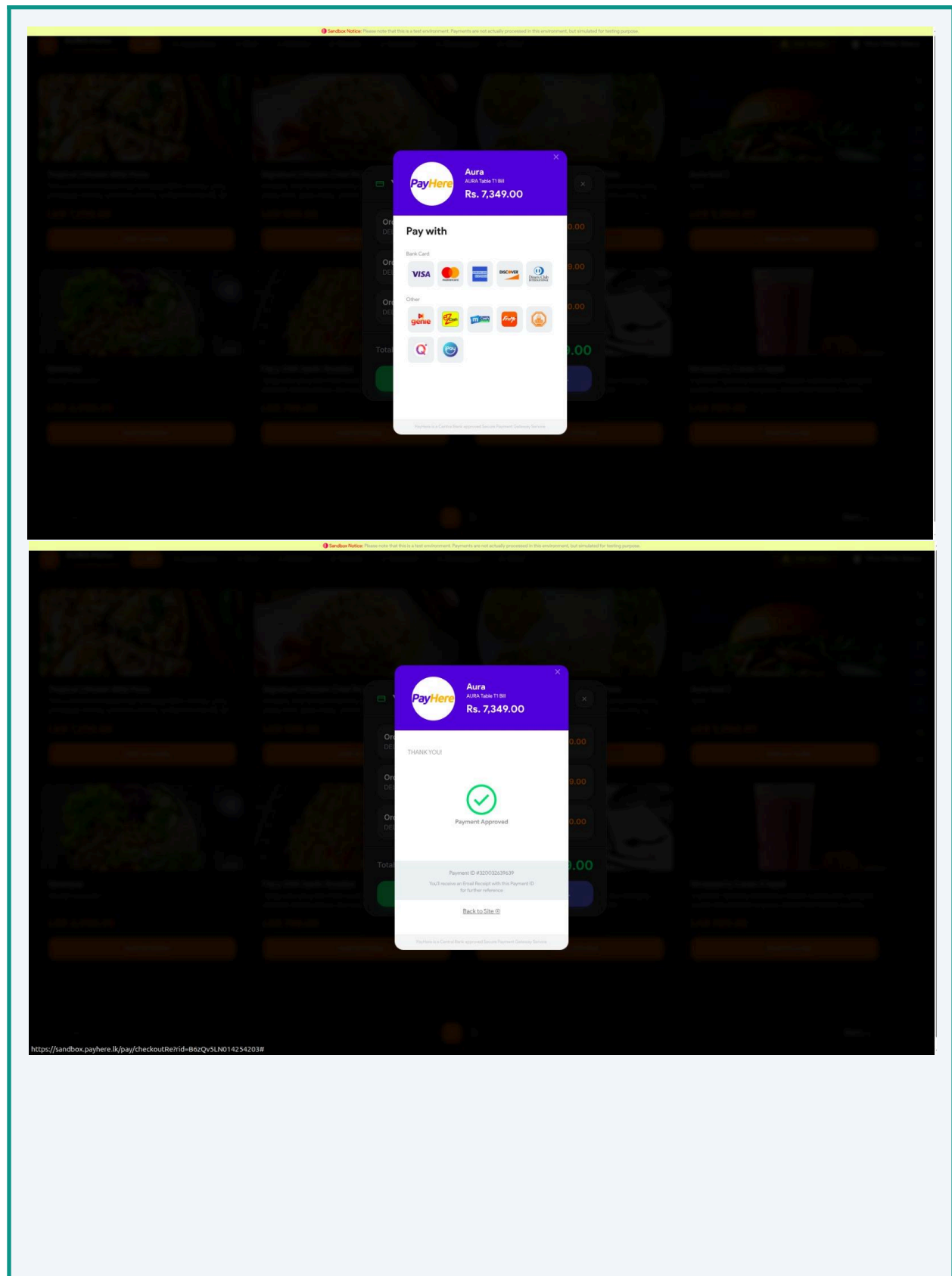
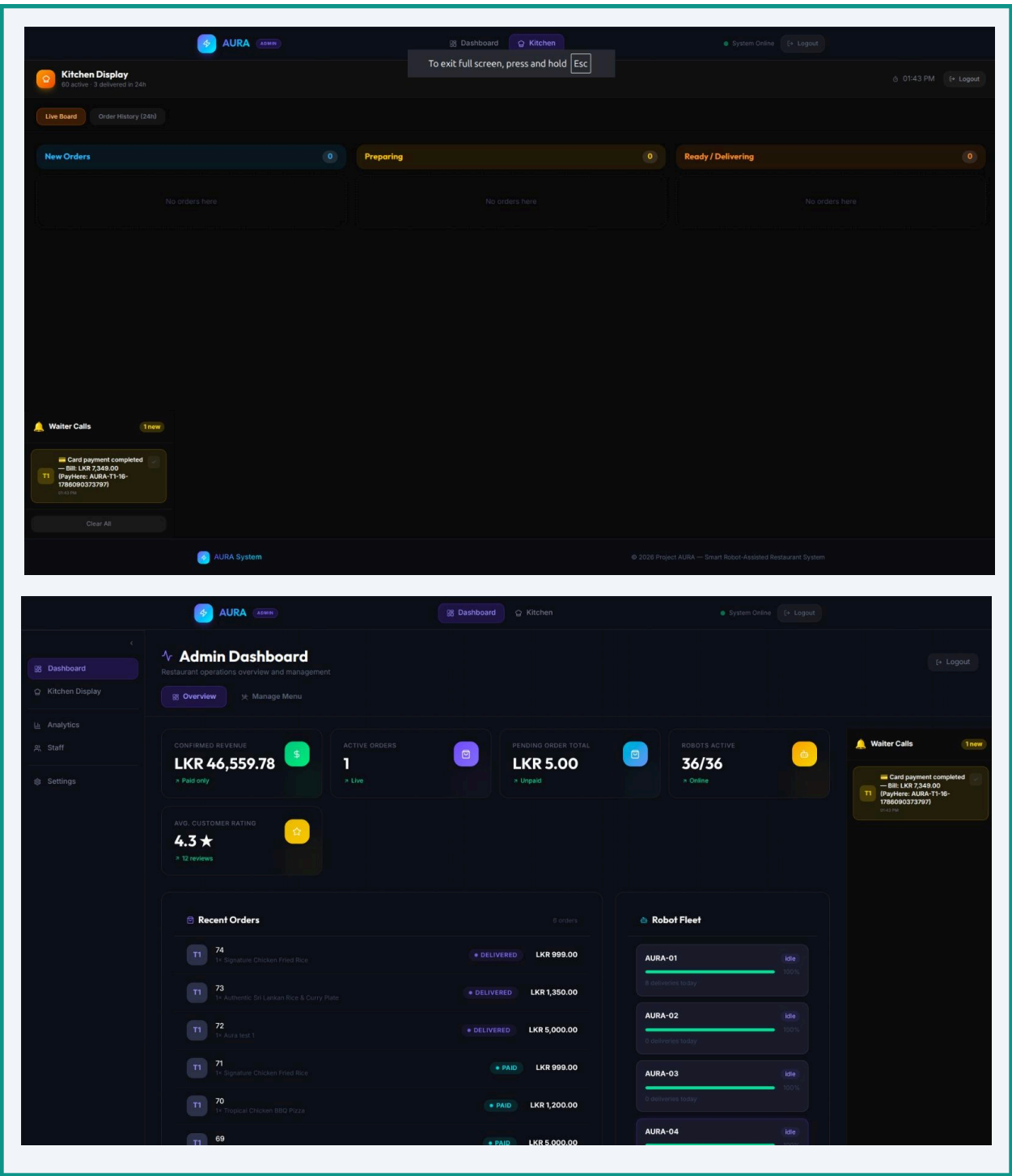
**Table-Side Payment via PayHere Gateway (Card/Cash) & Kitchen Notification**

Figure 10: Seamless settlement screen supporting Card or Cash payments via PayHere gateway with instant kitchen confirmation.







Analysis of Existing Products & Competitive Edge

To evaluate AURA's market positioning, a comparative analysis was conducted against traditional paper ordering, tablet/QR ordering systems, and high-cost walking delivery robots:

Feature / Metric	Traditional / QR Code	Walking Service Robots	AURA Robotic Assistant
Hardware Unit Cost	Low (LKR 10k-30k)	Extremely High (> \$3,500 - \$4,500)	Affordable (~ LKR 60000 out-of-pocket)
Customer Engagement	Passive / Monotonous	Moderate (Passing by)	High (Voice AI, Games, Selfies, Face Motion)
Infrastructure Requirements	Standard Wi-Fi	Wide aisles, smooth flat floors	Compact Tabletop, Plug-and-Play AC Power
Operational Integration	Order only (No physical AI)	Delivery only (No table AI)	Full Ecosystem (Order, Kitchen KDS, PayHere)

Target Market & Business Model Type

- **Primary Target Audience:** Urban restaurants, boutique cafés, themed dining venues, and family eateries looking to boost guest retention and streamline service.
- **Secondary Market:** High-end hotel dining rooms and casino restaurants seeking novel technological attractions.
- **Business Model Type:** B2B Hardware-Enabled Software-as-a-Service (HaaS + SaaS). Hardware sold/leased upfront with an ongoing monthly software maintenance subscription.

BUSINESS MODEL AND MARKETING PLAN

Overview of the Business Model

AURA operates a dual-stream revenue model: direct hardware deployment fees combined with recurring cloud subscription revenue. Restaurants purchase or lease the AURA tabletop hardware units and pay a monthly subscription for cloud API access (AWS hosting, Gemini Voice AI processing, menu synchronization, PayHere gateway maintenance, and continuous software updates).

Industry Context

The global restaurant automation and commercial service robotics market is projected to grow at over 16% CAGR. Rapid adoption of contactless ordering, increased labor costs, and consumer demand for experiential dining create a highly favorable market environment for accessible, table-based robotics.

Business Model Canvas

Canvas Block	Description & AURA Implementation
Value Proposition	Affordable, engaging tabletop robot combining voice AI, games, face-tracking motion, instant kitchen routing, and PayHere table-side payment.
Customer Segments	Mid-to-high tier urban restaurants, themed cafés, family diners, hotel food halls.
Channels & Revenue Streams	Direct B2B sales force, restaurant tech expos, website sales (aurarestaurant.tech). Hardware sales/leasing + Monthly SaaS subscription per active table.
Key Resources & Cost Structure	Robotic hardware designs, AWS cloud backend, Gemini AI integration, embedded Python/Spring Boot code. Component manufacturing, software maintenance, customer support.

Marketing Strategy and Analysis

- **Pilot Placement Program:** Provide trial installations to prominent urban dining establishments in Colombo and Kandy to build social proof and high-impact promotional media.
- **Experiential Social Marketing:** Leverage viral social media content generated by guests utilizing AURA's 'Selfie Mode' and interactive voice responses.
- **POS Integration Partnerships:** Partner with point-of-sale (POS) vendor networks and restaurant consultancy firms for bundled software/hardware distribution.

TECHNICAL OVERVIEW AND IMPLEMENTATION

Project Development Stage

Current Phase: PROTOTYPE DEVELOPMENT & SYSTEM INTEGRATION TESTING (TRL 6). The team has constructed fully operational physical hardware prototypes, executed full-stack cloud deployment on AWS, completed end-to-end integration across MQTT, PayHere gateway, and voice AI agents, and validated system reliability through rigorous testing.

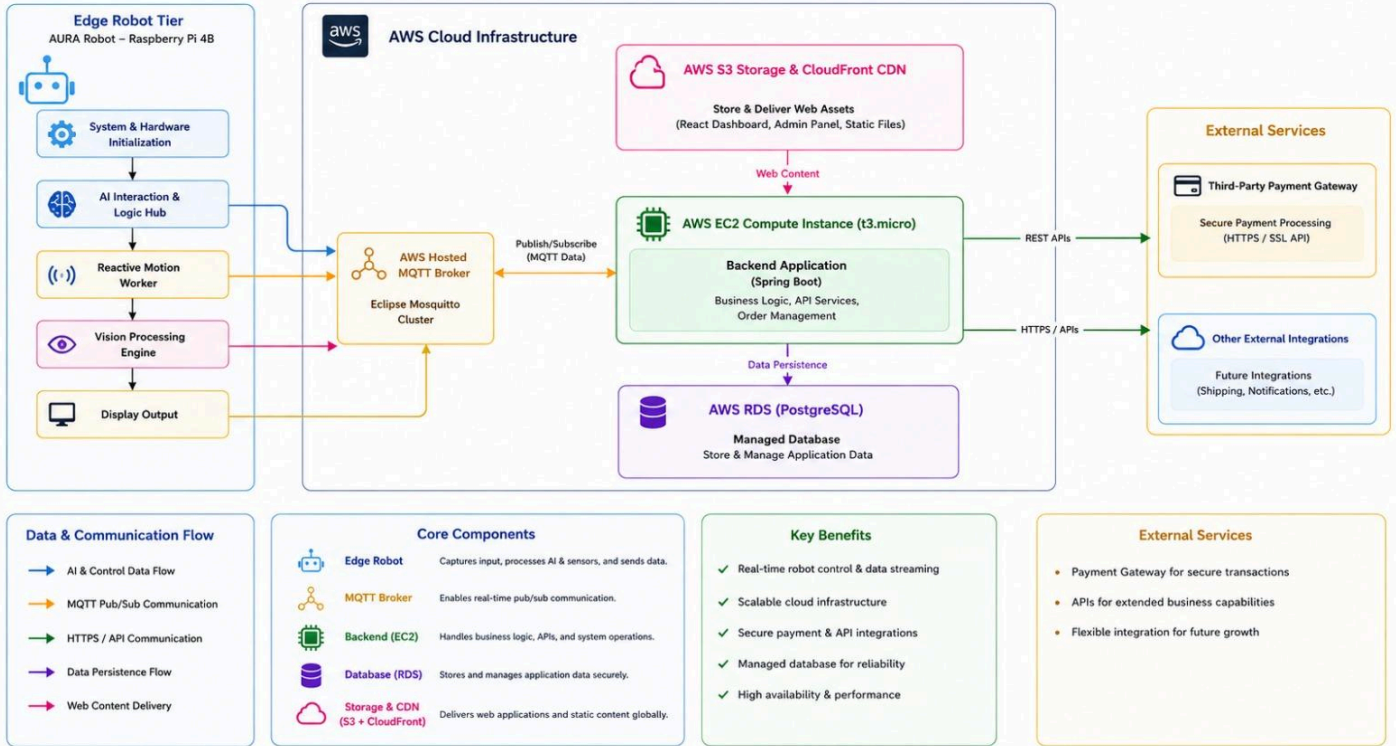
System Architecture & Architectural Diagrams

The AURA technical architecture comprises three core operational tiers: Edge Hardware, Cloud Backend, and Web User Interfaces. Below are the architectural schematics detailing data flows, component relationships, and deployment environments:

High-Level Architecture & Tec Stacks

Figure 11: Comprehensive data flow chart and AWS cloud architectural layout of the AURA robotic platform.

AURA Robot System – High-Level Architecture Diagram



Category	Technical Specifications
Backend	Java Spring Boot deployed via AWS Elastic Beanstalk / EC2 (Hibernate/JPA)
Database	AWS RDS (PostgreSQL Instance) (Cloud-managed, Dockerized dev mirroring)
Security	RBAC via JWT tokens, AWS Secrets Manager / .env isolation Amazon Cognito, AWS Secrets Manager, .env Isolation
Frontend	React + Vite + Tailwind CSS deployed via AWS S3 / CloudFront Amazon S3, Amazon CloudFront
UI / UX	WebSockets (Live Dashboard), Entertainment Module (Games/Songs)
Integrations	Gemini AI, External Payment Networks (Secure HTTPS/SSL over AWS Gateway) Amazon API Gateway

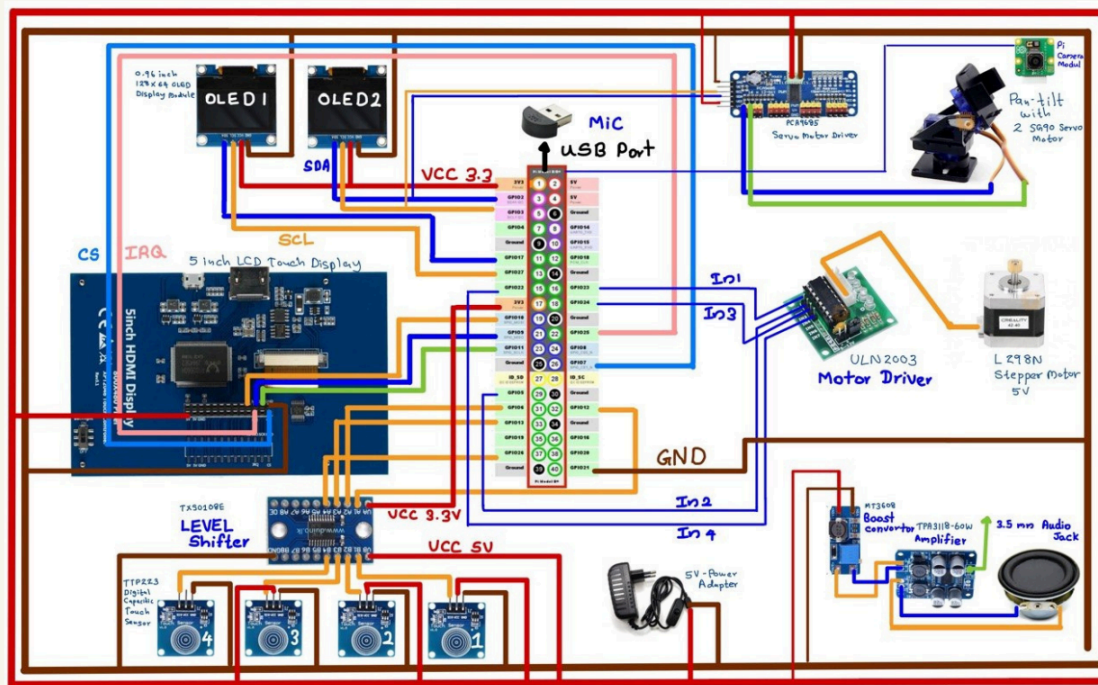
Software stack

	Category	 Technical Specifications
	Controller	 Raspberry Pi 4B (Multi-threaded Python, PWM)
	Sensors	 TTP223 Capacitive Touch  Pi Camera V2 (CSI)
	Actuators	 Dual SG90 Servos (PCA9685)  Dual 0.96" OLED displays
	Power & Cost	 TXB0108 Level Shifters Out-of-pocket: LKR 33,205.00
	Network	 AWS IoT Core / Hosted MQTT (Mosquitto on AWS EC2)  JSON over Wi-Fi

Hardware tech stack

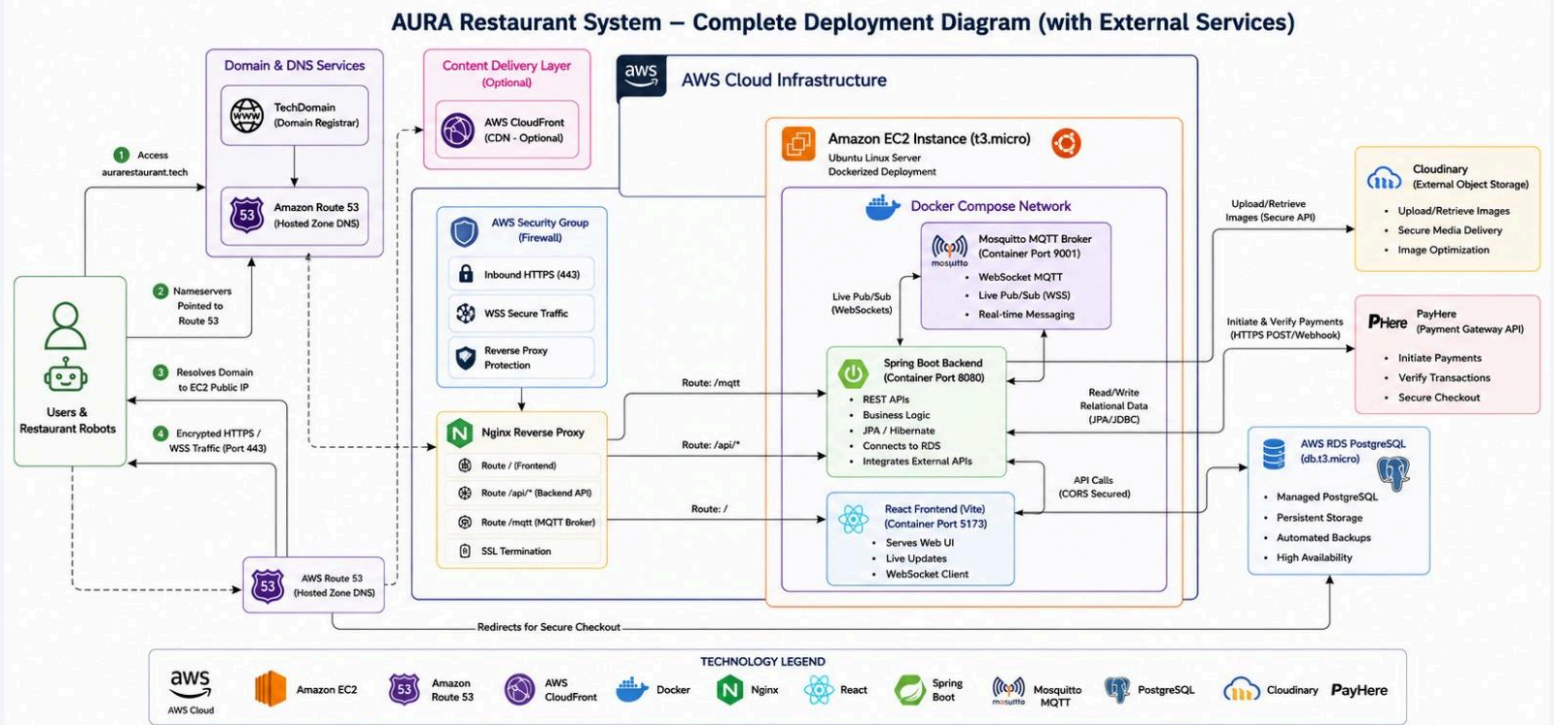
Detailed Edge Hardware Wiring & Circuit Diagram

Figure 12: Schematic of Raspberry Pi 4B hardware connections, servo drivers, dual OLED displays, amplifier, and touch sensors.



AWS Cloud Deployment & Infrastructure Network Diagram

Figure 13: Cloud infrastructure deployment diagram on AWS EC2, S3, Route 53, CloudFront, and PayHere Gateway integration.



Implementation Strategy & Technical Specifications

Hardware Engineering & Embedded Systems

- **Main Edge Controller:** Raspberry Pi 4B executing multi-threaded Python 3 engine for hardware control, PWM modulation, and cloud communication.
- **Vision Sensor:** Pi Camera V2 connected via CSI interface, driving real-time Computer Vision face detection.
- **Eye Displays:** Dual 0.96" I2C OLED screens representing animated, expressive eyes.
- **Main Interface:** 5.0-inch HDMI Capacitive Touch Display for menu navigation, games, and payment interaction.

- **Kinematics & Actuation:** Dual SG90 servo motors driven via PCA9685 I2C PWM controller for multi-axis neck tilting, coupled with a mechanical gear system driven by stepper/servo motors to physically rotate the robot around the table toward seated guests.
- **Plug-and-Play Power Supply:** Powered directly from standard café AC wall outlets via a step-down DC adapter (5V/3A), ensuring uninterrupted continuous operation without battery degradation.

Software Stack & Cloud Infrastructure

- **Backend Framework:** Java 17 Spring Boot RESTful API deployed via Docker Compose on an AWS EC2 instance (Ubuntu Linux).
- **Database Tier:** Managed AWS RDS (PostgreSQL) instance for transactional security and menu storage.
- **Frontend Stack:** ReactJS + Vite + Tailwind CSS hosted on AWS S3 and accelerated via AWS CloudFront CDN.
- **Real-Time Messaging:** Mosquitto MQTT Broker on AWS IoT Core / EC2 for real-time WebSocket communication between table robots, kitchen displays, and admin consoles.
- **Payment Gateway:** PayHere Payment Gateway REST API supporting secure Card transactions and cash settlement confirmation.
- **Voice AI Engine:** Google Gemini AI API integrated with speech-to-text/text-to-speech pipelines for conversational dining assistance.

Computer Vision & Kinematic Face Tracking

AURA utilizes OpenCV Haar Cascade classifiers on video streams captured by the Pi Camera V2. Upon detecting a customer's face within the camera frame, the system calculates x-y spatial coordinates and translates them into PWM step angles. These signals drive the dual SG90 neck servos and mechanical gear base motor, rotating the robot to directly face the customer, creating a responsive human-robot interaction.

FUTURE PROJECTIONS

Vision for the Future

AURA aims to become the benchmark smart dining assistant across South Asia and international hospitality markets, transforming traditional restaurants into automated, data-driven culinary spaces.

Growth Strategy

- **Phase 1 (Commercialization):** Deployment of pilot units across selected partner restaurants in Sri Lanka, followed by commercial rollouts in regional urban centers.
- **Phase 2 (Platform Scaling):** Development of advanced predictive analytics models for restaurant managers to forecast dish demand and streamline inventory purchasing.
- **Phase 3 (Ecosystem Expansion):** Expanding hardware options to include specialized modular attachments (e.g., thermal plate heaters, digital menu stands).

Milestones

- **Short-Term (0–3 Months):** Finalize physical prototype, execute AWS deployment, integrate PayHere payment gateway, and perform stress testing. (Status: COMPLETED)
- **Medium-Term (3–12 Months):** Deploy pilot units in 5 commercial restaurants, gather guest feedback, refine AI voice models, and initiate commercial sales.
- **Long-Term (12–36 Months):** Scale to 100+ active restaurant installations, establish contract manufacturing, and expand sales to regional markets.

FINANCIAL PROJECTIONS

Revenue Model

- **Hardware Sales:** Out-of-pocket production cost ~ LKR 58,700 per prototype unit. Wholesale/retail pricing model targets LKR 85,000 – 110,000 per unit.
- **SaaS Subscription:** Monthly software subscription per table (LKR 3,500 – 5,000/month) covering AWS hosting, Gemini AI API credits, maintenance, and menu updates.

Funding Requirements & Out-of-Pocket Hardware Cost

The prototype unit was engineered with extreme cost efficiency, achieving an out-of-pocket hardware build cost of LKR 58,700.00. Below is the detailed component cost breakdown:

Category	Components	Cost(LKR)
Controller & Compute	Raspberry Pi 4B (4GB), MicroSD Card, USB Card Reader	19,100.00
Sensors & Camera	Pi Camera V2 (CSI), TTP223 Capacitive Touch, Digital Touch Sensors	5,500.00
Actuators & Displays	Dual MG90 Servos, PCA9685 Driver, Dual 0.96" OLEDs, 5" Touch Screen, HDMI Cables	16,500.00
Power & Circuitry	TXB0108 Level Shifters, 5V 3A DC Power Adapter, Jumper Wires, PCB Board, Barrel Connector	2,500.00
Enclosure & Mechanics	3D Printer Filament (PLA+), FDM Printing Hours, Servo Brackets, Gears	8,500.00
Audio Interface	USB Microphone Dongle, I2S Speaker Module, 4 Ohm 5W Speaker	2,500.00
Accessories	Box Stickers, User Manual, Solder Wire, Glue, Chip 32	4,100.00
Total Out-of-Pocket Cost	Complete Physical & Electronic Prototype	LKR 58,700.00

Financial Forecasts (3-Year Projection)

- **Year 1:** Deploy 25 units across 3 partner venues. Annual Projected Revenue: LKR 3.5M. Operating Margin: 26%.
- **Year 2:** Expand to 120 deployed units across 15 venue locations. Annual Projected Revenue: LKR 16.7M. Operating Margin: 33%.
- **Year 3:** Scale to 400 deployed units regionally. Annual Projected Revenue: LKR 55.6M. Operating Margin: 35%.

Risk Analysis & Mitigation

- **Supply Chain Bottlenecks:** Mitigated by utilizing local industrial 3D printing and standardized off-the-shelf electronic components.
- **Network / Cyber Security Risks:** Mitigated by implementing reverse-proxy Nginx security, JWT authentication, and SSL/TLS encryption across all AWS communication.

SOCIAL IMPACT AND SUSTAINABILITY

Social Impact

- **Staff Workload Reduction:** Automating repetitive order-taking allows existing service staff to focus on high-value hospitality duties, food presentation, and customer support.
- **Inclusive Dining Experience:** Provides an engaging, non-intimidating interactive technology experience accessible to diners of all age groups.
- **Local Innovation Boost:** Supplies high-tech automation solutions engineered locally by Sri Lankan engineering students, fostering local technology commercialization.

Environmental Sustainability

- **Eco-Friendly 3D Fabrication:** Robotic chassis components fabricated using recyclable PLA bioplastics derived from renewable organic sources.
 - **Low Power Plug-and-Play Setup:** Operating on a direct AC grid plug-and-play transformer eliminates frequent battery replacements and electronic battery disposal waste.
 - **Paperless Operations:** Digital menus and table-side digital receipts significantly reduce physical paper usage in restaurant dining rooms.
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ROADMAP AND IMPLEMENTATION TIMELINE

Development Roadmap & Timeline Table

Phase / Duration	Key Tasks & Deliverables	Status
Phase 1 (Months 1-2)	Requirements gathering, 3D chassis design, component sourcing, initial Spring Boot & React architecture setup.	Completed
Phase 2 (Months 3-4)	Hardware assembly, Pi Camera OpenCV face tracking, Dual OLED eye animation, MQTT broker setup.	Completed
Phase 3 (Months 5-6)	AWS Cloud deployment, PayHere gateway payment integration, Gemini Voice AI tuning, full system integration testing.	Completed
Phase 4 (Months 7-8)	Pilot trial deployment in local partner venue, user feedback collection, performance optimization.	In Progress
Phase 5 (Months 9-12)	Commercial marketing launch, scale production, expand SaaS sales team across urban centers.	Planned

TEAM DETAILS AND EXPERTISE

Team ReLife consists of dedicated undergraduate engineering students from the Faculty of Engineering, University of Peradeniya, bringing together expertise across full-stack software development, cloud engineering, embedded hardware systems, mechanical design, and artificial intelligence.

Member 1

STUDENT PHOTO	
FULL NAME	Sahabandu Kankanam Arachchige Kaveesha Madhushan
NIC	200118803147
EMAIL ADDRESS	e21245@eng.pdn.ac.lk / kaveeshamadhushan1776@gmail.com
CONTACT NO	076 0609159
PROJECT CONTRIBUTION	Managed AWS EC2/S3/RDS cloud infrastructure, Docker containerization, Route 53 DNS routing, Nginx reverse proxy configuration, and MQTT messaging pipeline. Build the React Frontends and Main Reservation page

Member 2

STUDENT PHOTO	
FULL NAME	Suraweera Gamage Induwara Amaranga
NIC	200213301576
EMAIL ADDRESS	e21024@eng.pdn.ac.lk
CONTACT NO	076 0537926
PROJECT CONTRIBUTION	Designed 3D chassis models in Blender/CAD, orchestrated additive manufacturing 3D printing, engineered mechanical gear rotation system, and assembled physical robotic body.

Member 3

STUDENT PHOTO	
FULL NAME	Hewa Gajaman Kankanamge Venuri Dulya Chamuduni Dissanayake
NIC	200263600680
EMAIL ADDRESS	e21113@eng.pdn.ac.lk
CONTACT NO	070 3130243
PROJECT CONTRIBUTION	Developed Light/Dark theme user interfaces, integrated Gemini Voice AI agent pipeline, designed table-side mini-games, and conducted usability testing.

Member 4

STUDENT PHOTO	
FULL NAME	Thennakoon Mudiyanse Induranga Chamikara Thennakoon
NIC	200202502419
EMAIL ADDRESS	e21407@eng.pdn.ac.lk
CONTACT NO	071 9637096
PROJECT CONTRIBUTION	Engineered Spring Boot backend, React interface, Raspberry Pi Python edge controller, PayHere payment integration, and Computer Vision face tracking algorithms.